CASE STUDY 04 · CLOUD & DEVOPS

Linux sysadmin automation with Bash

A set of modular Bash scripts that turn repetitive server chores — user access, environment setup, process monitoring, log parsing and health checks — into single, repeatable commands.

ROLE
Sole developer

STACK
Linux · Bash · Git

FOCUS
Operations automation

STATUS
Complete

01 / why i built it

Why I built it

Doing the same server tasks by hand is slow and error-prone: a missed step when creating a user, a forgotten environment variable, a log nobody reads until something breaks. The DevOps idea I wanted to practice is simple — **if you do it twice, script it**.

02 / what the scripts do

What the scripts do

User access management

Create, modify and remove users and their permissions consistently.

Environment setup

Set up environment variables and configuration the same way every time.

Process monitoring

Check that important processes are running and report the ones that aren't.

Log parsing

Scan logs automatically and pull out the lines that matter.

Health checks

Routine checks of system health to streamline maintenance.

Version control

Scripts and configuration documentation managed in Git and GitHub.

03 / design

Design

Each script does one job and can be run on its own or combined with the others. Keeping them modular made them easier to test, reuse and document — the same principle behind larger automation and infrastructure-as-code tools.

The foundation came from my Operating Systems course, where I worked in Ubuntu from the command line throughout: files and directories, processes, threads, concurrency and scheduling.

04 / what i learned

What I learned

- Writing Bash that is safe to run more than once.
- Thinking in terms of repeatable operations instead of one-off commands.
- Documenting operational work so someone else can run it.

This is the base I'm building on for DevOps: next come CI/CD pipelines, containers and infrastructure as code.